

1 Nowcasting Framework, Model Construction, and Empirical Results

1.1 Overview of the Nowcasting System

This chapter presents the final pseudo real-time nowcasting system for Armenia’s real GDP growth. The purpose of the framework is to estimate quarterly year-on-year GDP growth before the official release by combining quarterly macroeconomic information, monthly official indicators, and faster market- and search-based signals. The empirical design is organized around three information stages:

- **Early:** only month-1 fast indicators and lagged macroeconomic information are available;
- **Mid:** first monthly official releases for the quarter become available;
- **Late:** the within-quarter information set is close to complete before the GDP publication.

The stage structure matters because the information set changes materially within the quarter. In normal periods, relatively standard bridge and regularized regression models are often sufficient. In crisis periods, especially in the *Early* stage, the same-quarter signal is sparse and noisy. For that reason, the final system does not rely on a single specification. Instead, it combines structural benchmarks, machine-learning models, and targeted regime adjustments.

This point is especially important given the focus of the thesis on alternative data sources. In the Armenian application, alternative indicators are not an ornamental addition to an already complete official dataset. They are the main same-quarter signals available at the beginning of the quarter, and they remain valuable later as rapid indicators of external stress, regime change, and turning points that may not yet be visible in the official releases.

1.2 Data Construction and Feature Engineering

The target variable is Armenia’s quarterly real GDP growth in year-on-year terms, denoted by

$$y_t = 100 \times \frac{GDP_t - GDP_{t-4}}{GDP_{t-4}}. \quad (1)$$

The explanatory information set contains four main data blocks:

1. **Quarterly macroeconomic variables:** Russia’s GDP growth, CPI, REER, employment, unemployment, income-related variables, and exchange-rate indicators.

2. **Monthly official variables:** CPI, Economic Activity Index, industrial growth, construction growth, services growth, monetary aggregates, loans, deposits, and remittance indicators.
3. **High-frequency fast variables:** AMD/USD, AMD/RUB, Brent oil price, copper price, and curated Google Trends blocks.
4. **Shock and external-leading features:** composite search-based shock measures, Russia-linked interaction terms, and financial-stress-style exchange-rate indicators.

1.2.1 Detailed Description of the Data Used

The empirical dataset used in this study combines quarterly macroeconomic series, monthly official indicators, and fast-moving external and search-based proxies. Building this panel was itself a substantial part of the research, since Armenia does not have a ready-made public nowcasting database.

Quarterly block. The quarterly block contains the target series together with slower-moving macroeconomic controls. It includes Armenia’s real GDP growth, Russia’s real GDP growth, CPI, REER, employment growth, unemployment, labor-income variables, transfer-income variables, and expenditure-side aggregates such as consumption, investment, exports, and imports. This block anchors the system in medium-run macroeconomic structure and provides stability when same-quarter information is still thin.

Monthly official block. The monthly official block contains the main real-sector and financial indicators used in bridge-style nowcasting. The final panel includes CPI, the Economic Activity Index, industrial growth, construction growth, services growth, interest rates, cash in circulation, monetary aggregates, deposits, loans, and remittance indicators. It also incorporates monthly extensions collected from the Central Bank of Armenia and ArmStat. These variables carry much of the forecasting power in the *Mid* and *Late* stages.

Fast block. The fast block was introduced to capture information before official monthly releases appear. It includes AMD/USD, AMD/RUB, Brent oil, copper prices, and several groups of Google Trends indicators. These variables are particularly important in the *Early* stage, when no same-quarter official monthly indicators are yet available.

Alternative-data and shock block. Because the thesis is specifically about GDP nowcasting with alternative data sources, this block is central to the empirical strategy. Rather than using individual search terms directly, the framework aggregates them into thematic composites: housing stress, relocation intensity, banking stress, and jobs/IT

relocation. These are then combined into an overall shock composite. This reduces dimensionality and makes the resulting signals easier to interpret economically. In substantive terms, the block provides rapid evidence on stress, mobility, financial anxiety, and relocation-related demand before those developments become visible in the official monthly statistics.

External-leading block. Because Armenian GDP is highly exposed to external disturbances, especially those transmitted through Russia, the framework also includes interaction variables such as oil–RUB, copper–RUB, RUB stress, and a simple financial-stress proxy derived from AMD/USD and AMD/RUB volatility. These variables help the model react earlier to turning points originating outside Armenia.

Quarterly engineered features were constructed to reflect Armenia’s external dependence and crisis sensitivity. In particular, the final framework included:

$$\text{MigrationInflux}_t = \frac{\max(0, 100 - FX_{yoy,t}^{AMD/USD}) \cdot \max(0, PI_t^{yoy})}{100}, \quad (2)$$

$$\text{RU_ARM_Gap}_t = y_t - y_t^{RU}, \quad (3)$$

$$\text{RU_Oil_Link}_t = y_t^{RU} \cdot \text{Oil}_t, \quad (4)$$

$$\text{RU_AMD_RUB_Link}_t = y_t^{RU} \cdot FX_t^{AMD/RUB}. \quad (5)$$

At the monthly level, the fast block was aggregated into quarterly-stage features using means, last-observation summaries, and a simple Almon-style weighted average:

$$x_t^{ALMON} = 0.2119x_{t,1} + 0.5761x_{t,2} + 0.2119x_{t,3}, \quad (6)$$

where the weights were adapted to the number of months available at each stage.

Google Trends variables were not used individually in the final preferred system. Instead, they were grouped into interpretable composite blocks:

- FAST_SHOCK_HOUSING,
- FAST_SHOCK_RELOCATION,
- FAST_SHOCK_BANKING,
- FAST_SHOCK_JOBS_IT,
- FAST_SHOCK_COMPOSITE.

In addition, external-leading features were added in order to capture Armenia’s sen-

sitivity to Russian and global shocks:

$$\text{RUS_LINK_OIL_RUB}_m = FX_m^{AMD/RUB} \cdot Oil_m, \quad (7)$$

$$\text{RUS_LINK_COPPER_RUB}_m = FX_m^{AMD/RUB} \cdot Copper_m, \quad (8)$$

$$\text{RUS_LINK_RUB_STRESS}_m = |\Delta FX_m^{AMD/RUB}|, \quad (9)$$

$$\text{FIN_STRESS_PROXY}_m = \frac{|\Delta FX_m^{AMD/USD}| + |\Delta FX_m^{AMD/RUB}|}{2}. \quad (10)$$

This data structure reflects the basic economic logic of the Armenian nowcasting problem. GDP is quarterly and released with delay, but the relevant signals arrive at different frequencies and through different channels. The framework therefore combines slow-moving macroeconomic structure, monthly official activity indicators, and fast market and search proxies in a single leakage-safe design.

The importance of alternative data also changes across stages. In *Early*, the nowcast relies heavily on lagged quarterly structure plus month-1 alternative indicators such as exchange rates, commodity prices, Google Trends composites, and external stress features because same-quarter official monthly releases are not yet available. In *Mid*, those alternative indicators remain in the information set, but they are joined by the first official monthly releases, which sharply improve identification of current-quarter activity. In *Late*, the official block becomes much richer, yet the alternative-data block still matters because it helps monitor external transmission channels and detect abrupt changes that may not be fully summarized by the official averages alone.

1.3 Pseudo Real-Time Evaluation Design

All models were evaluated in an expanding-window pseudo real-time backtest. Let t denote the target quarter. For each quarter and each information stage, the model was estimated using only the information available at that point:

- in the **Early** stage, the information set contains lagged quarterly macro variables together with month-1 alternative indicators from exchange rates, commodity prices, Google Trends composites, and shock features, while no same-quarter official monthly variables are used;
- in the **Mid** stage, month-2 alternative indicators remain available and are joined by the first official monthly release for the target quarter;
- in the **Late** stage, the within-quarter information set is close to complete, so the nowcast uses a much richer official monthly block while retaining the alternative indicators for external and regime monitoring.

In practice, this means that preprocessing, feature selection, scaling, imputation, PCA reduction, and model fitting were all re-estimated inside each training window. This limits information leakage and makes the reported forecast errors methodologically credible.

For each model and stage, the following evaluation metrics were computed:

$$MAE = \frac{1}{T} \sum_{t=1}^T |\hat{y}_t - y_t|, \quad (11)$$

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2}, \quad (12)$$

$$MAPE = \frac{100}{T} \sum_{t=1}^T \left| \frac{\hat{y}_t - y_t}{y_t} \right|. \quad (13)$$

Empirical prediction intervals were also constructed from the rolling residual distribution. For each stage-model pair, 50% and 90% bands were estimated from historical absolute-residual quantiles.

1.4 Benchmark Models

1.4.1 Autoregressive Baseline (AR)

The autoregressive benchmark uses lagged GDP growth and current-quarter regime dummies:

$$y_t = \alpha + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_4 y_{t-4} + \gamma' D_t + \varepsilon_t, \quad (14)$$

where D_t includes crisis and seasonal indicators. This model serves as the minimum benchmark and is especially useful in extreme quarters because it is stable and difficult to overfit.

1.4.2 Bridge Regression

The bridge model links quarterly GDP growth to quarterly macro variables and selected monthly activity indicators. The operational specification is estimated as a ridge regression:

$$\hat{\beta}^{ridge} = \arg \min_{\beta} \left\{ \sum_{t=1}^T (y_t - X_t \beta)^2 + \lambda \|\beta\|_2^2 \right\}. \quad (15)$$

This model is useful because it balances interpretability and robustness while drawing directly on activity indicators such as the Economic Activity Index, industrial growth, construction, and services.

1.4.3 Elastic Net

Elastic Net is used as the main regularized machine-learning benchmark:

$$\hat{\beta}^{EN} = \arg \min_{\beta} \left\{ \sum_{t=1}^T (y_t - X_t \beta)^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right] \right\}. \quad (16)$$

It is especially useful in the *Early* stage, where it can screen a high-dimensional feature set and retain the most informative fast indicators without producing an unstable saturated regression.

1.4.4 Factor-Augmented PCA Regression

The factor-augmented PCA benchmark is an approximate factor model. Monthly variables are first standardized and reduced by principal components:

$$X_t^{(m)} \approx F_t \Lambda', \quad (17)$$

where F_t denotes the retained principal components. GDP growth is then forecast using ridge regression on the quarterly block and the latent factors:

$$y_t = \alpha + Z_t' \delta + F_t' \theta + \varepsilon_t. \quad (18)$$

This serves as a useful comparison benchmark, but it is not a true ragged-edge state-space DFM.

1.4.5 Dynamic Factor Model (DFM)

The true structural benchmark in the thesis is a mixed-frequency dynamic factor model estimated with `DynamicFactorMQ`. The measurement equations can be written as

$$x_{i,m} = \lambda_i f_m + u_{i,m}, \quad (19)$$

$$y_t = \lambda_y f_t^{(q)} + u_{y,t}, \quad (20)$$

where $x_{i,m}$ are monthly observed indicators, f_m is the latent common factor, and $f_t^{(q)}$ is the quarterly aggregation of the latent monthly factor. The factor evolves according to

$$f_m = \phi f_{m-1} + \eta_m. \quad (21)$$

The baseline DFM uses one latent factor, first-order factor dynamics, and optionally AR(1) idiosyncratic components. In the context of this thesis, it is the most defensible theory-grounded mixed-frequency benchmark because it handles ragged-edge monthly information directly.

1.4.6 DFMSHockAdjusted

The raw DFM still tends to overpredict in sharp collapse quarters because the latent factor smooths abrupt regime shifts. Therefore, a crisis-corrected version was introduced:

$$\hat{y}_t^{adj} = (1 - \omega_t)\hat{y}_t^{DFM} + \omega_t s_t, \quad (22)$$

where s_t is the current-quarter official activity signal derived from the Economic Activity Index, industrial growth, construction, and services, and ω_t increases when the gap between the DFM nowcast and the contemporaneous activity signal becomes large. The goal is to preserve the structural role of the DFM while correcting its most important crisis-quarter bias.

1.4.7 Random Forest

The Random Forest benchmark approximates the conditional expectation function non-linearly:

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^B T_b(X_t), \quad (23)$$

where $T_b(\cdot)$ denotes the b -th regression tree. This model can capture nonlinear interactions among fast indicators and crisis dummies, but in a small sample it can be less stable than regularized linear alternatives.

1.4.8 Gradient Boosting

The boosting benchmark iteratively updates the forecast function by fitting trees to the residuals:

$$f_M(x) = \sum_{m=1}^M \nu h_m(x), \quad (24)$$

where $h_m(\cdot)$ are shallow trees and ν is the learning rate. This model is intended to capture nonlinear relationships and interactions, especially in the *Mid* stage.

1.4.9 Huber Regression

Huber regression minimizes a robust loss:

$$L_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq \delta, \\ \delta \left(|r| - \frac{1}{2}\delta \right), & |r| > \delta. \end{cases} \quad (25)$$

In principle, this should help when forecast errors are heavy-tailed. In practice, it remained unstable in the Armenian GDP application and therefore served mainly as a diagnostic benchmark.

1.4.10 EarlyShockBridge

This benchmark was designed for the *Early* stage only. It uses a restricted information set composed of lagged GDP, fast shock composites, exchange rates, oil, copper, and crisis dummies:

$$y_t = \alpha + \beta' z_t + \varepsilon_t, \quad (26)$$

where z_t contains only crisis-sensitive fast indicators. The model is estimated using a Huber-type regression in order to reduce sensitivity to isolated large errors.

1.4.11 Shadow Model

The Shadow model is a stability benchmark using only a very small set of robust predictors. It can be written as

$$y_t = \alpha + \beta' x_t^{shadow} + \varepsilon_t, \quad (27)$$

where x_t^{shadow} includes only a handful of stable variables such as lagged GDP, Russian GDP, CPI, exchange-rate information, and the financial stress proxy. The Shadow model is not designed to win on average accuracy; its value lies in indicating when richer specifications may be drifting too far from historically plausible values.

1.4.12 Combination Models

Three combination models were used:

1. **SimpleEnsemble**: arithmetic average of a subset of machine-learning models.
2. **AdaptiveEnsemble**: weighted average of recent model predictions, where the weights are inversely proportional to recent regime-specific absolute errors:

$$w_{j,t} \propto \frac{1}{MAE_{j,t-1:t-k} + \epsilon}, \quad \hat{y}_t = \sum_j w_{j,t} \hat{y}_{j,t}. \quad (28)$$

3. **ShockSwitch**: a regime-dependent combination of Bridge, Elastic Net, and Random Forest forecasts:

$$\hat{y}_t = (1 - p_t) \hat{y}_t^{normal} + p_t \hat{y}_t^{shock}, \quad (29)$$

where $p_t \in [0, 1]$ is the estimated probability of being in a shock regime.

1.4.13 EarlyShockAdjusted

The final targeted month-1 crisis benchmark is **EarlyShockAdjusted**. It starts from the best *Early*-stage baseline and then applies a smooth crisis correction:

$$\hat{y}_t^{ESA} = \hat{y}_t^{base} - A_t, \quad (30)$$

where the adjustment A_t depends on a lockdown dummy, a smoothed shock-regime probability, agreement across multiple fast indicators, and the intensity of the shock composites. In the most extreme lockdown case, the model also uses the AR benchmark as an anchor in order to avoid implausibly high nowcasts.

1.5 Shock-Regime Classification

One of the more important features of the final framework is that crisis adjustment is not based on a hard threshold alone. Instead, a logistic regression classifier is used to estimate the probability of entering a shock regime:

$$p_t = \Pr(S_t = 1 \mid z_t) = \frac{1}{1 + e^{-(\alpha + z_t'\beta)}}, \quad (31)$$

where z_t includes the fast shock composite, banking and housing shock blocks, AMD/USD, AMD/RUB, oil, copper, and external stress features.

Alongside the probability model, a sign-agreement rule was imposed. A downward crisis correction is activated only when several unrelated fast indicators jointly signal distress. This reduces false alarms in non-crisis quarters while preserving responsiveness during genuine systemic shocks.

1.6 Empirical Results

1.6.1 Data, Information Stages, and Model Architecture Summary

Before turning to the headline forecast results, it is useful to summarize the empirical ingredients of the system in a compact way. The following tables provide that overview and make the chapter easier to read as a self-contained account of the nowcasting framework.

Table 1: Main data blocks used in the Armenian GDP nowcasting system, with emphasis on alternative sources.

Data block	Frequency	Main examples	Main role in the system
Quarterly macro block	Quarterly	Armenia GDP growth, Russia GDP growth, CPI, REER, employment, unemployment, income and expenditure aggregates	Provides slow-moving macro structure and stabilizes nowcasts when same-quarter information is incomplete.
Monthly official block	Monthly	Economic Activity Index, CPI, industry, construction, services, loans, deposits, remittances, monetary aggregates	Provides the core real-activity signal in the <i>Mid</i> and <i>Late</i> stages once official releases begin to arrive.
Alternative market block	Daily/monthly aggregated	AMD/USD, AMD/RUB, Brent oil, copper prices, financial stress proxy	Provides the first same-quarter signals in <i>Early</i> and remains useful for monitoring external transmission.
Alternative search and shock block	High-frequency aggregated	Housing stress, relocation, banking stress, jobs/IT relocation, external stress interactions	Captures regime change, crisis intensity, and behavioral shifts before they are visible in the official data.

Table 2: Information available in each nowcasting stage and the role of alternative data.

Stage	Information available	Main forecasting difficulty	Preferred interpretation
Early	Lagged quarterly macro variables; month-1 alternative data from exchange rates, commodity prices, Google Trends composites, and shock features; no same-quarter official monthly releases	Sparse same-quarter information and maximum exposure to abrupt crisis noise	Alternative data has its highest marginal value here; best stage for regularized or explicitly shock-adjusted models.
Mid	Lagged quarterly structure; month-1 and month-2 alternative indicators; first official monthly releases for activity, prices, remittances, and selected financial variables	Combining heterogeneous signals without overweighting noise	Best stage for adaptive combinations, because alternative data and first official releases must be read jointly.
Late	Lagged quarterly structure; near-complete official monthly block for the target quarter; alternative indicators retained for external and shock monitoring	Refining the point nowcast and quantifying residual uncertainty	Best stage for the final operational nowcast, where official data dominate but alternative data still help confirm or question the signal.

Table 2 makes the role of alternative data explicit. In the *Early* stage, the nowcast would be materially weaker without exchange rates, commodity prices, Google Trends composites, and shock indicators because there is almost no same-quarter official information to work with. In the *Mid* stage, alternative indicators do not disappear; instead, they complement the first official releases and help determine whether the official signal is consistent with broader market and behavioral movements. In the *Late* stage, the official block becomes more informative, but alternative data remain useful as a fast check on external spillovers and abrupt changes in regime.

Table 3: Model taxonomy used in the thesis.

Family	Models	Mathematical core	Main role in the thesis
Time-series / structural	AR , Bridge , FactorAugmentedPCA , DFM	Lag dynamics, ridge bridge equations, latent-factor extraction, mixed-frequency state-space modeling	Provides interpretable benchmarks and the main theory-grounded representation of ragged-edge nowcasting.
Regularized ML	ElasticNet , Huber , EarlyShockBridge	Penalized regression, robust loss, crisis-sensitive restricted regressions	Screens high-dimensional predictors and stabilizes sparse-information forecasting.
Nonlinear ML	RandomForest , GradientBoosting	Tree ensembles and nonlinear interaction learning	Tests whether nonlinearities improve performance in a small-sample open-economy setting.
Combination models	SimpleEnsemble , AdaptiveEnsemble , ShockSwitch	Forecast averaging, inverse-error weighting, regime-dependent switching	Delivers the main operational nowcast by combining complementary model classes.
Shock- adjusted models	DFMShockAdjusted , EarlyShockAdjusted	Structural nowcast plus crisis correction; baseline nowcast minus shock adjustment	Corrects the most important crisis-quarter failures of the baseline models.

Table 4: Shock-adjusted model variants and their economic logic.

Baseline model	Adjusted version	Adjustment logic	Main use case
DFM	DFMShockAdjusted	The raw factor nowcast is pulled toward a contemporaneous activity signal when the factor model diverges too far in a shock quarter.	Crisis-sensitive correction for <i>Mid</i> and <i>Late</i> stage factor nowcasts.
Best <i>Early</i> baseline	EarlyShockAdjusted	Subtracts both crisis adjustment and lockdown- and shock-probability-driven correction from the month-1 baseline nowcast.	Dedicated repair for extreme sparse-information crisis quarters, especially 2020 Q2 <i>Early</i> .
Static model combination	ShockSwitch	Forecast weights shift toward a shock specification when the estimated crisis probability rises.	Regime-aware operational averaging under changing conditions.

1.6.2 Headline Accuracy

The final expanding-window backtest produced the following best-performing models by stage:

- **Early:** `EarlyShockAdjusted`, MAPE = 2.658%;
- **Mid:** `AdaptiveEnsemble`, MAPE = 2.708%;
- **Late:** `AdaptiveEnsemble`, MAPE = 2.504%.

The main implication is that no single model dominates in every information stage. The *Early* stage is the most fragile because it combines sparse information with the greatest exposure to abrupt crisis noise. In that setting, the targeted `EarlyShockAdjusted` benchmark performs best. Once the quarter progresses and official monthly releases begin to arrive, the adaptive combination of several models becomes superior.

Table 5: Headline benchmark accuracy by nowcast stage (MAPE, %).

Model	Early	Mid	Late
ElasticNet	2.696	3.009	2.697
AdaptiveEnsemble	2.891	2.708	2.504
Bridge	3.327	3.356	3.165
ShockSwitch	3.093	3.139	2.907
EarlyShockAdjusted	2.658	–	–
DFMShockAdjusted	3.858	3.476	3.156
DFM	3.858	4.110	3.781
AR	3.115	3.245	3.170

Table 5 contains the central ranking of the chapter. Three patterns stand out. First, `EarlyShockAdjusted` is the best *Early*-stage specification, which indicates that crisis-sensitive correction materially improves month-1 nowcasts. Second, among models available in all three stages, `AdaptiveEnsemble` is the best operational model, with an average MAPE of roughly 2.70%. Third, the structural factor benchmarks remain important for interpretation, but they are not the best performers in average-accuracy terms.

Table 6: Top five model rankings by information stage.

Rank	Early model	MAPE	Mid model	MAPE	Late model	MAPE
1	EarlyShockAdjusted	2.658	AdaptiveEnsemble	2.708	AdaptiveEnsemble	2.504
2	ElasticNet	2.696	ElasticNet	3.009	Shadow	2.506
3	AdaptiveEnsemble	2.891	RandomForest	3.125	ElasticNet	2.697
4	Shadow	3.006	ShockSwitch	3.139	ShockSwitch	2.907
5	ShockSwitch	3.093	SimpleEnsemble	3.146	RandomForest	3.060

Table 7: Average accuracy by model family and stage (MAPE, %).

Family	Early	Mid	Late
Combination	3.540	2.998	2.862
Structural	3.906	3.892	3.572
ML	4.327	4.139	3.392

Table 8: Selected robustness diagnostics for key benchmark models.

Model	Stage	Overall MAPE	Shock MAPE	Non-shock MAPE	Prediction bias
EarlyShockAdjusted	Early	2.658	2.702	2.602	1.150
ElasticNet	Early	2.696	2.960	2.364	-0.157
AdaptiveEnsemble	Mid	2.708	2.720	2.683	0.555
AdaptiveEnsemble	Late	2.504	2.580	2.324	0.310
DFMShockAdjusted	Mid	3.476	3.179	4.132	-0.380
DFMShockAdjusted	Late	3.156	3.135	3.205	0.206
DFM	Mid	4.110	4.100	4.132	-3.458
AR	Early	3.115	3.594	2.511	-1.576

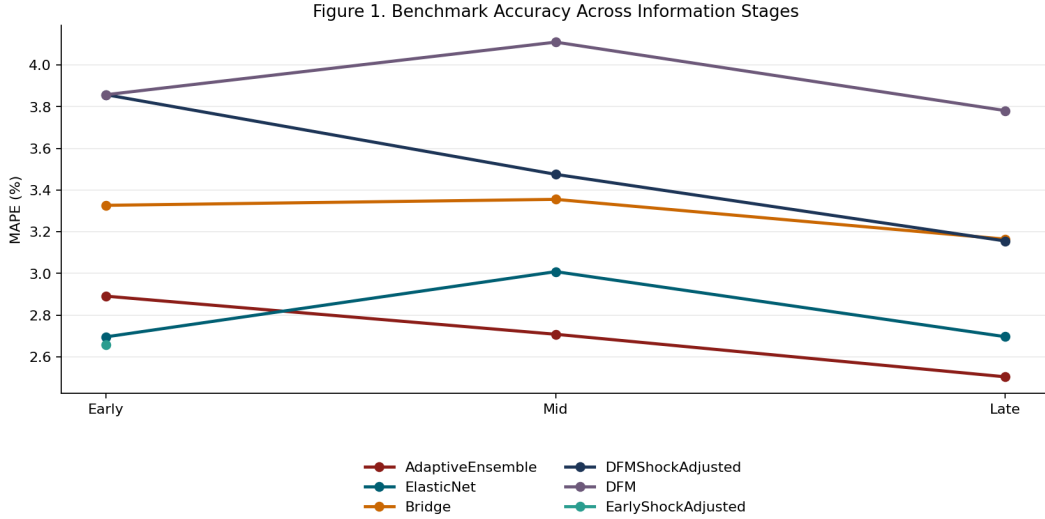


Figure 1: Benchmark accuracy across the three information stages. The figure shows that model ranking is stage-dependent: regularized linear models dominate in the sparse-information *Early* stage, while combination models dominate once the information set becomes richer.

Figure 1 summarizes the empirical results visually. It makes clear that the Armenian GDP nowcasting problem does not look the same throughout the quarter. In the *Early* stage, the forecaster sees only a limited fast-information set and the value of explicit shock correction is highest. In the *Mid* and *Late* stages, the informational advantage shifts toward models that combine several complementary signals.

1.6.3 Structural and Crisis Benchmarks

Among factor-based and structural benchmarks, the hierarchy is the following:

- **DFM** is the main theory-grounded mixed-frequency benchmark;
- **DFMShockAdjusted** is the strongest crisis-corrected factor benchmark;
- **FactorAugmentedPCA** underperforms both because it compresses the monthly information too early.

The **DFMShockAdjusted** benchmark achieves:

- **Mid**: $\text{MAPE} = 3.476\%$;
- **Late**: $\text{MAPE} = 3.156\%$.

This confirms that the true DFM remains academically valuable, but the operationally best model in this application is an adaptive ensemble rather than a single structural specification.

Table 9: Factor and structural benchmark comparison (MAPE, %).

Model	Early	Mid	Late
AR	3.115	3.245	3.170
Bridge	3.327	3.356	3.165
FactorAugmentedPCA	5.372	5.273	4.587
DFM	3.858	4.110	3.781
DFMShockAdjusted	3.858	3.476	3.156

Table 9 shows why it is necessary to distinguish between structural relevance and operational accuracy. The PCA-factor benchmark performs clearly worse because it compresses the monthly information too early. The true mixed-frequency DFM is methodologically superior, but it still needs a crisis correction. Once that correction is added, **DFMShockAdjusted** becomes the strongest factor benchmark, especially in *Mid* and *Late*.

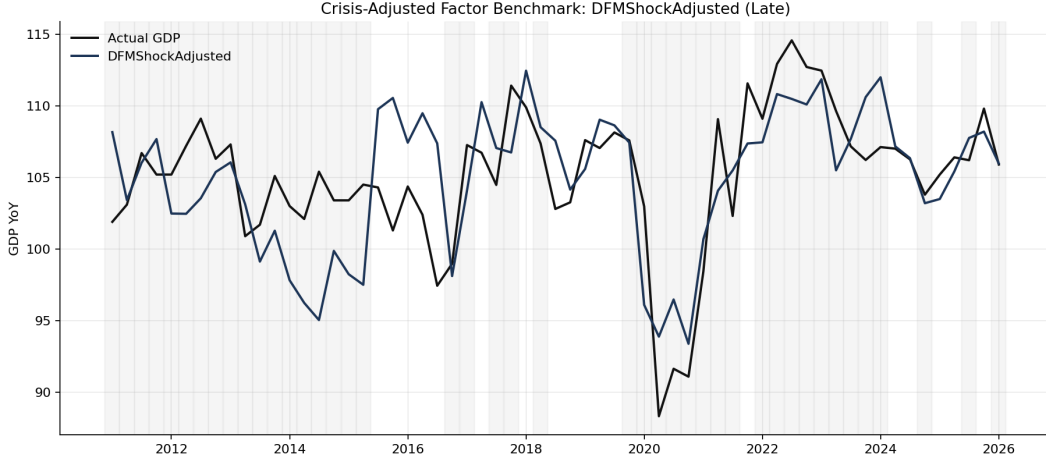


Figure 2: Actual GDP growth and the crisis-adjusted factor benchmark in the *Late* stage. The figure illustrates the role of the DFM as a structural benchmark and shows how the crisis-adjusted version reduces the raw factor model’s overprediction during abrupt downturns.

Figure 2 matters for the structural interpretation of the results. The adjusted DFM is not the best model in average-accuracy terms, but it is the strongest theory-grounded factor benchmark. The figure shows that the state-space model becomes substantially more useful once the contemporaneous activity signal is allowed to discipline the latent factor during crisis quarters.

1.6.4 Crisis-Quarter Results

The hardest quarter in the sample is 2020 Q2, especially in the *Early* stage, where same-quarter official monthly indicators are not yet available. After the final lockdown-specific calibration, the dedicated `EarlyShockAdjusted` benchmark reduced the error for **2020 Q2 Early** to

$$\text{APE}_{2020Q2, \text{ Early}}^{\text{ESA}} = 1.872\%. \quad (32)$$

This result matters because earlier general-purpose models produced much larger errors for the same case. The implication is not that one universal model solves every state of the world. Rather, the evidence shows that:

1. general-purpose models perform well on average;
2. the DFM is the correct structural mixed-frequency benchmark;
3. targeted regime adjustments are necessary for extreme month-1 crisis quarters.

Table 10: Selected crisis-quarter absolute percentage errors (%). Abbreviations: AdaEns = **AdaptiveEnsemble**, ENet = **ElasticNet**, DFM-SA = **DFMShockAdjusted**, ESA = **EarlyShockAdjusted**.

Quarter	Stage	AdaEns	ENet	Bridge	DFM-SA	ESA
2020 Q2	Early	18.007	18.045	15.034	21.423	1.872
2020 Q2	Mid	13.501	12.156	13.115	7.412	–
2020 Q2	Late	13.661	15.875	14.041	6.285	–
2021 Q2	Early	7.590	7.420	8.349	2.628	7.420
2021 Q4	Early	7.885	7.576	7.128	7.084	8.397
2022 Q2	Mid	5.601	5.732	6.131	0.517	–
2022 Q2	Late	4.457	4.931	4.410	1.862	–

Table 10 is the main stress-test table in the chapter. It shows that the hardest crisis case is not simply “a recession quarter” but specifically the *Early* stage of 2020 Q2. In that setting, the dedicated **EarlyShockAdjusted** benchmark dominates the alternatives by a very large margin. By contrast, in *Mid* and *Late*, the most important crisis correction comes from **DFMShockAdjusted**. This supports the stage-dependent design of the final system.

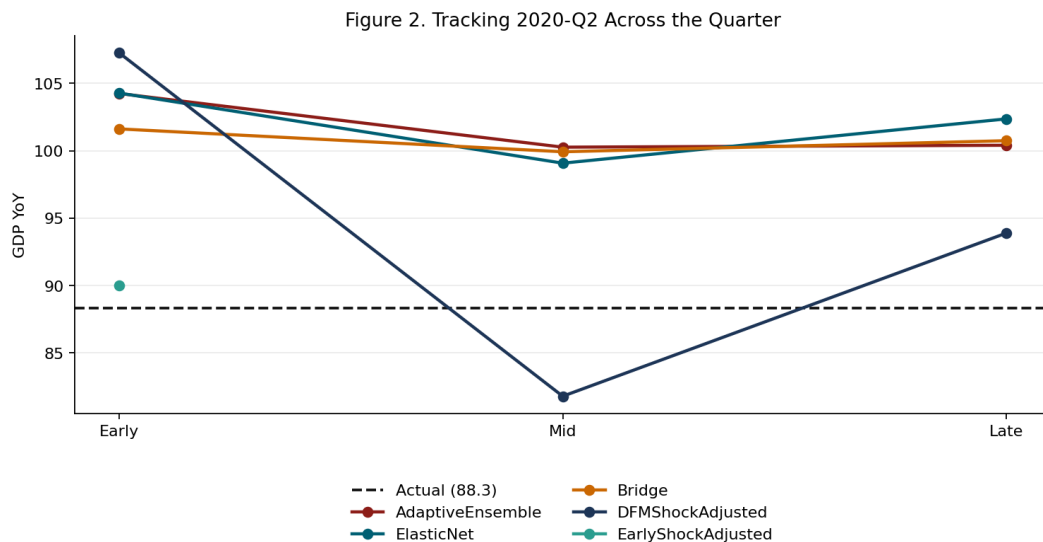


Figure 3: Evolution of the 2020 Q2 nowcast across the quarter. This is the central crisis-quarter figure in the thesis because it shows how forecast accuracy changes between *Early*, *Mid*, and *Late* as new information arrives.

Figure 3 functions as the main case study of the chapter. It shows that the hard part of crisis nowcasting is not simply forecasting a recession. The real difficulty is forecasting a sudden collapse with only month-1 information. Once official monthly releases begin to appear, the forecast improves quickly. That is why the final conclusion is stage-dependent rather than universally model-dependent.

1.6.5 Preferred Final Model Interpretation

From an *operational* perspective, the preferred final model is **AdaptiveEnsemble**, because it is the most accurate specification among models that can be used in all three information stages and because it is strongest in *Mid* and *Late*. From a *structural* perspective, the preferred benchmark is **DFM**. For *extreme early-shock correction*, the relevant benchmark is **EarlyShockAdjusted**. This stage- and regime-dependent conclusion is one of the main contributions of the thesis.

1.6.6 Detailed Model-by-Model Interpretation

Average ranking alone is not sufficient. The chapter also needs to explain why the main models work, where they break down, and what those patterns imply for practical now-casting.

Table 11: Interpretation of the main benchmark models.

Model	Where it performs well	Where it fails or weakens	Empirical interpretation
AdaptiveEnsemble	Best full-stage operational model; strongest in <i>Mid</i> and <i>Late</i> ; low average error and limited bias.	Still weak in the most extreme month-1 collapse, especially 2020 Q2 <i>Early</i> .	Combination across model classes is the most reliable operational strategy once multiple signals become available.
ElasticNet	Strong in <i>Early</i> ; good non-shock performance; benefits from regularization when the predictor set is large and sparse.	Less dominant once richer official information becomes available; still vulnerable in severe shock quarters.	Regularization is highly valuable when the information set is thin, but not sufficient by itself for crisis repair.
Bridge	Economically interpretable and reasonably stable across normal quarters.	Does not dominate in any stage; weaker than the leading ML and combination models in both average and crisis performance.	The bridge model remains useful as an interpretable benchmark, but not as the preferred operational system.
DFM	Strongest theory-grounded mixed-frequency benchmark; academically defensible structure.	Systematic overprediction, especially in <i>Mid</i> ; weak crisis handling because the latent factor smooths abrupt breaks.	Structural rigor alone is not enough in a shock-prone small open economy.
DFMShockAdjusted	Clear improvement over raw DFM in <i>Mid</i> and <i>Late</i> ; strongest factor benchmark in crisis-sensitive settings.	Provides no meaningful improvement in <i>Early</i> ; still not the best average-accuracy model overall.	Crisis correction materially improves the structural benchmark once contemporaneous activity information exists.
EarlyShockAdjusted	Best <i>Early</i> -stage model; decisive gain in 2020 Q2 <i>Early</i> ; strongest targeted repair for sparse-information crisis nowcasting.	Available only for the <i>Early</i> stage and not intended as the universal final model.	A dedicated shock-adjusted month-1 benchmark is necessary for extreme crisis quarters.
AR	Stable baseline; relatively good post-2022 performance; useful anchor against implausible forecasts.	Persistent overprediction in shock periods and weaker use of contemporaneous information.	Simplicity gives robustness, but at the cost of responsiveness.
ShockSwitch	Balanced regime-aware combination with respectable rankings in all stages.	Does not beat AdaptiveEnsemble on average and does not solve the deepest early-COVID miss by itself.	Regime switching helps, but flexible adaptive averaging remains stronger operationally.

The time-series and structural models matter mainly because they define the economically interpretable baseline of the thesis. The **AR** specification is stable and continues to perform reasonably well in the later post-2022 period, but its negative mean residual shows persistent overprediction in shock episodes. The **Bridge** model is easier to explain economically than most machine-learning benchmarks, yet it remains weaker than the best operational models. The raw **DFM** is the clearest example of the tradeoff between academic structure and empirical flexibility: in theory it is the correct mixed-frequency benchmark, but in practice its residual profile shows that it smooths through sharp collapses too aggressively and therefore overpredicts when Armenia experiences abrupt turning points.

The machine-learning results are clearly stage-dependent. **ElasticNet** is strong because the *Early* nowcast is a sparse-information problem, and penalization helps identify the most useful fast indicators without producing an unstable saturated regression. The nonlinear ML models, especially **RandomForest** and **GradientBoosting**, do not dominate the benchmark tables, which is consistent with the small-sample nature of the Armenian GDP application. Their flexibility is not enough to offset limited historical depth and abrupt regime changes. The **Huber** specification likewise remains diagnostic rather than operational, because robust loss on its own does not solve the information-availability problem.

The combination and shock-adjusted models represent the strongest practical contribution of the thesis. **AdaptiveEnsemble** is the best operational model because it combines structural and ML signals without committing to a single specification across all regimes. **ShockSwitch** is a useful regime-aware combination, but it does not outperform the adaptive ensemble. Most importantly, the shock-adjusted models show where targeted correction is actually needed. **DFMShockAdjusted** repairs the structural factor benchmark once *Mid* and *Late* information becomes available, while **EarlyShockAdjusted** is the key month-1 crisis benchmark that addresses the most difficult failure in the sample.

Figure 11. Bias Profile of Selected Models by Stage

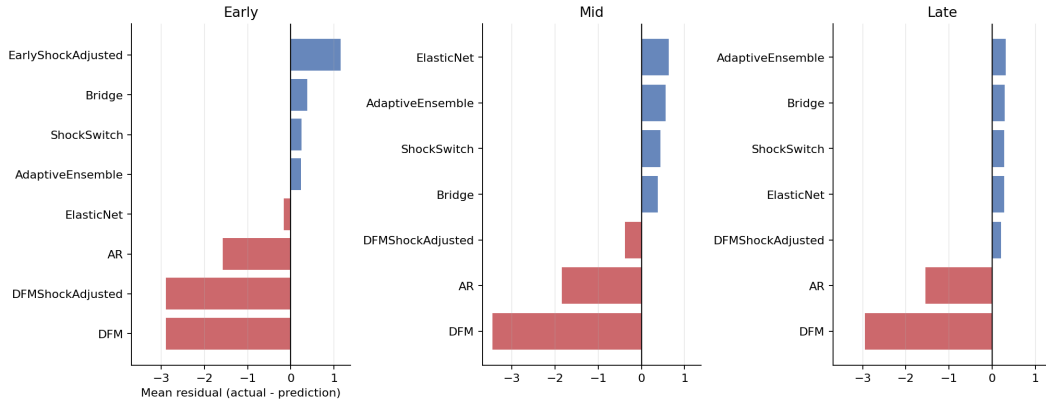


Figure 4: Bias profile of selected benchmark models by stage. Positive residuals indicate underprediction and negative residuals indicate overprediction because the residual is defined as actual minus prediction. The figure is useful for identifying which models fail because they are too optimistic and which fail because they react too slowly to new information.

Figure 4 helps interpret the ranking tables more carefully. The raw **DFM** and **AR** benchmarks show the strongest negative residuals, meaning that they tend to overpredict GDP during difficult episodes. By contrast, **AdaptiveEnsemble** and **ElasticNet** stay much closer to zero bias, while **DFMShockAdjusted** visibly reduces the structural overprediction problem in *Mid* and *Late*.

1.6.7 Forecast Revision Dynamics

Average error metrics do not fully describe how the forecasting system behaves over time. It is also useful to examine the full cloud of nowcasts as it evolves across quarters. The spaghetti chart below does that by plotting many model-stage nowcasts as thin lines and then highlighting the ensemble mean, the preferred operational model, and the realized GDP path.

Two points stand out in this figure. First, in normal quarters the highlighted nowcasts cluster fairly tightly around realized GDP, which implies limited model disagreement when the macroeconomic environment is stable. Second, in shock quarters the fan of predictions widens considerably, especially around 2020 Q2 and the later Russia-related adjustment period. This confirms that the largest nowcast errors arise when the information set is sparse and the economy is undergoing abrupt change. In other words, the Armenian nowcasting problem is both stage-dependent and regime-dependent.

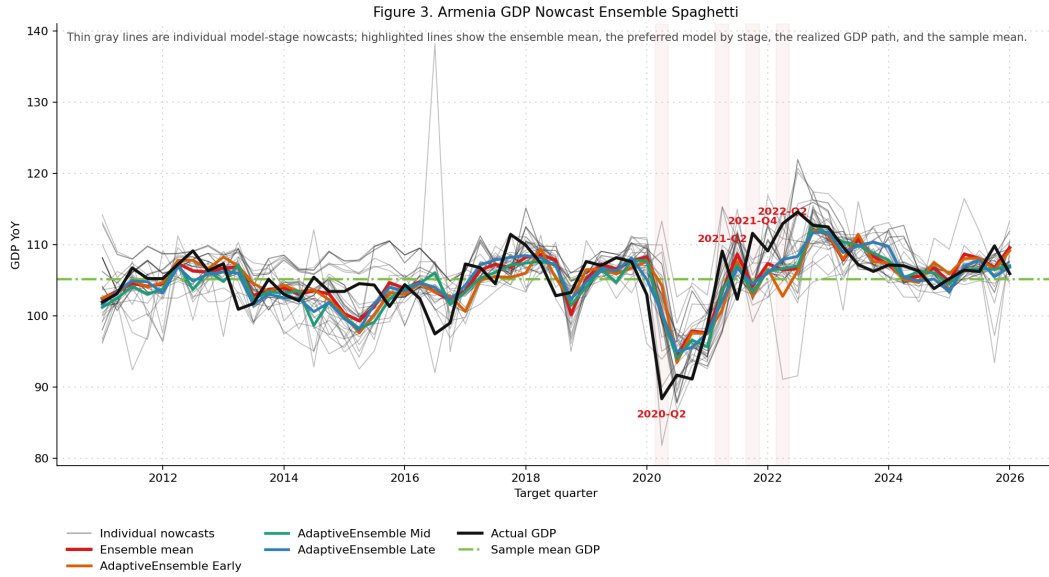


Figure 5: Ensemble-style spaghetti chart for Armenian GDP nowcasting. Thin gray lines show individual model-stage nowcasts across target quarters, while the highlighted lines show the ensemble mean, the **AdaptiveEnsemble** paths by information stage, the realized GDP path, and the sample mean. The figure illustrates both the dispersion of model-based nowcasts and the way forecasts cluster around the realized GDP trajectory outside the most difficult shock quarters.

Figure 5 adds an ensemble perspective to the chapter. The model's average performance is not simply the result of one favorable stage or one unusual quarter. Instead, the chart shows the broader cloud of nowcasts produced by competing specifications and highlights how the preferred operational model evolves relative to the ensemble mean and the realized GDP path. At the same time, the wider divergence around quarters such as 2020 Q2 makes clear why targeted early-shock correction remains necessary.

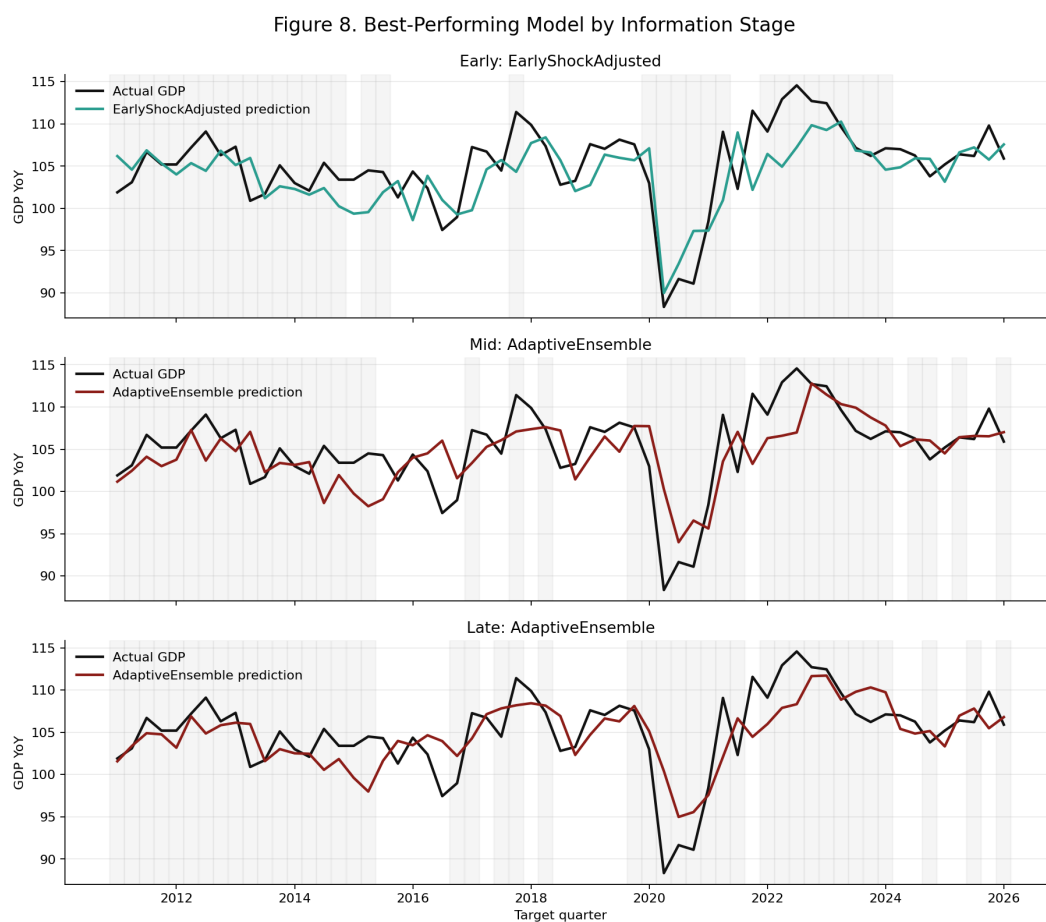


Figure 6: Best-performing model by stage across the full sample. The figure shows the realized GDP path against the winning specification in *Early*, *Mid*, and *Late*. This makes the stage-dependent nature of the final system visually explicit.

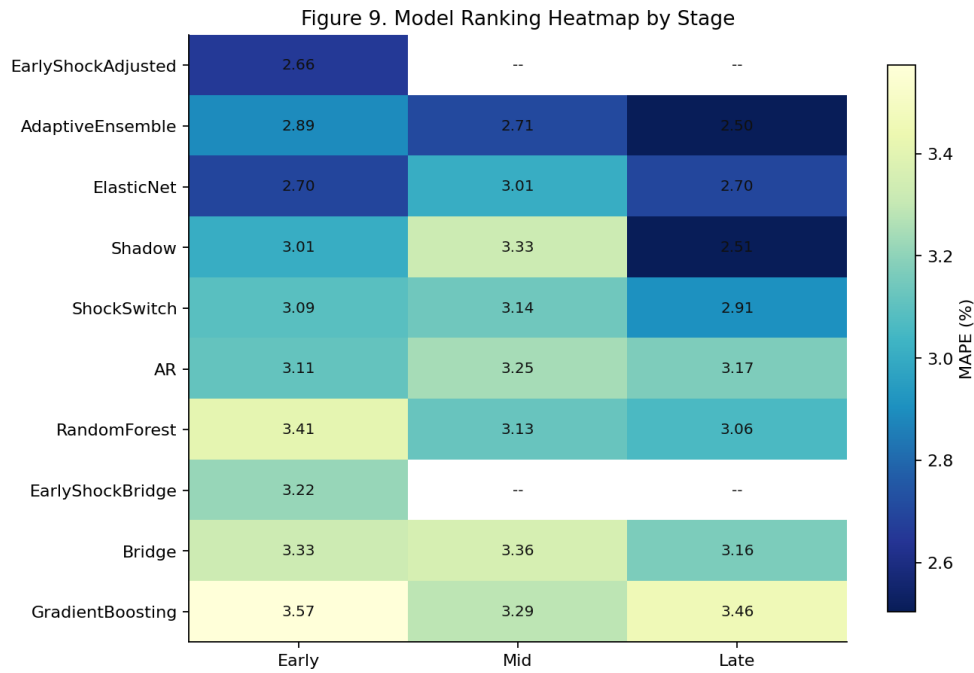


Figure 7: Heatmap of model ranking by information stage. Lower MAPE values indicate stronger performance. The visual compactly shows that model dominance changes across stages and that no single baseline wins everywhere.

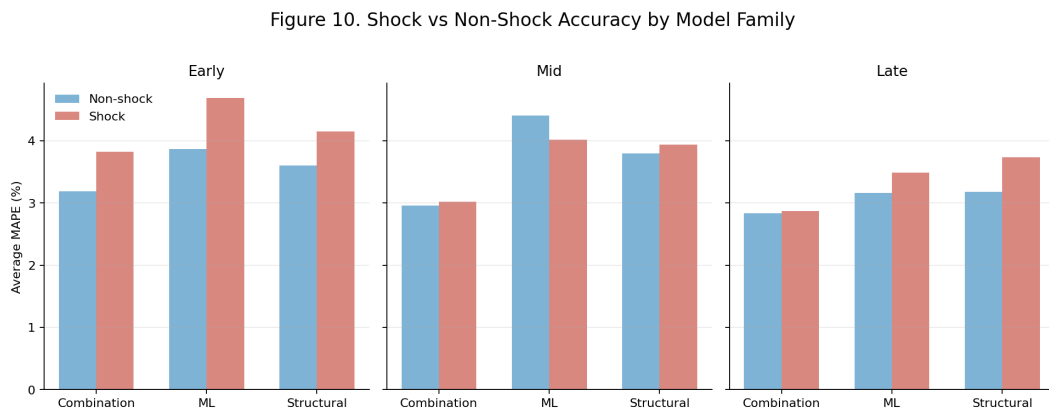


Figure 8: Shock and non-shock accuracy by model family. The figure highlights that combination models are the most robust family overall, while structural and pure ML families are more stage-dependent under crisis conditions.

The remaining visual diagnostics complement the main tables by showing information availability, uncertainty, family-level behavior, and the direct tracking performance of the preferred operational model.

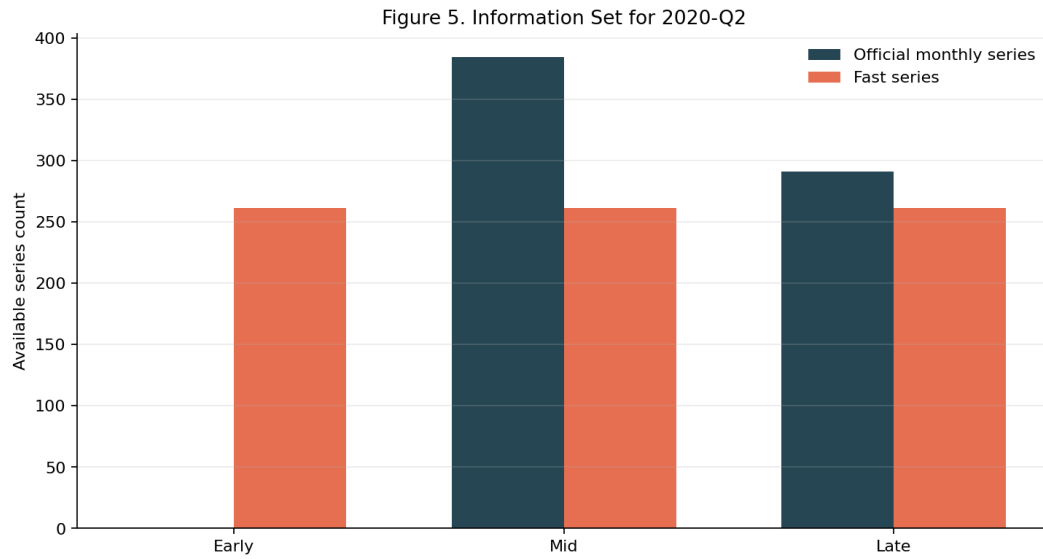
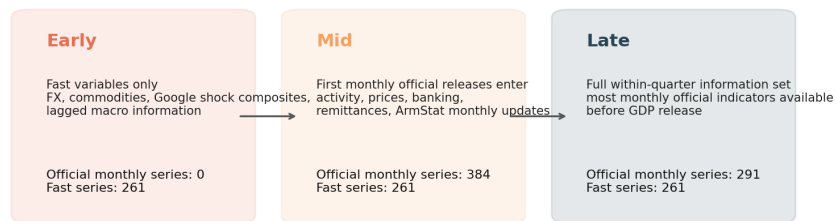


Figure 9: Information-set composition for 2020 Q2. The *Early* stage contains no same-quarter official monthly series, whereas the *Mid* and *Late* stages progressively incorporate richer official information.

Figure 6. Release Calendar and Information Flow Across the Quarter



The information set broadens from fast indicators in Early to fuller monthly official releases in Late, which is why crisis nowcasts improve as the quarter progresses.

Figure 10: Release calendar and information flow across the quarter. The figure illustrates the methodological logic of the framework: the *Early* stage relies mainly on fast indicators, while the *Late* stage benefits from a far richer official monthly information set.

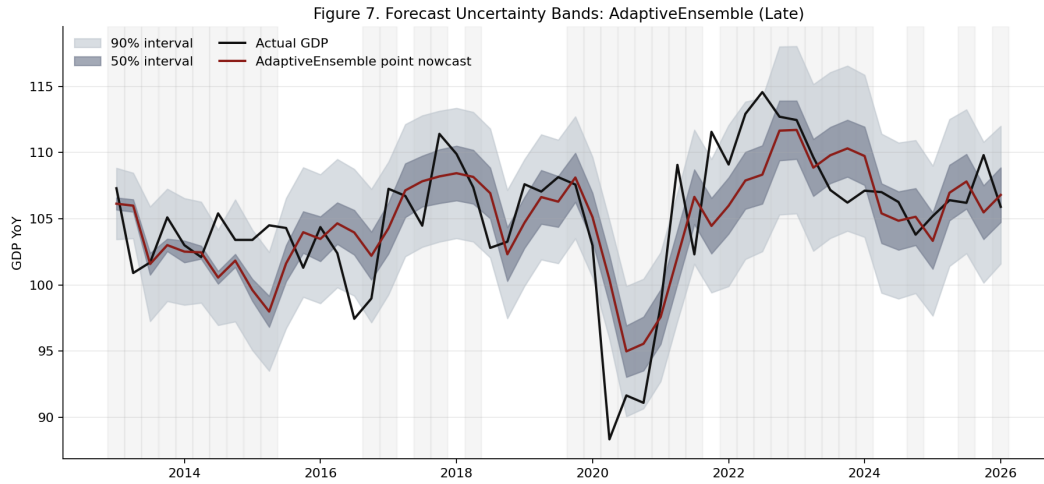


Figure 11: Forecast uncertainty bands for the preferred late-stage model. The 50% and 90% empirical intervals show that the framework produces not only point nowcasts but also uncertainty assessments.

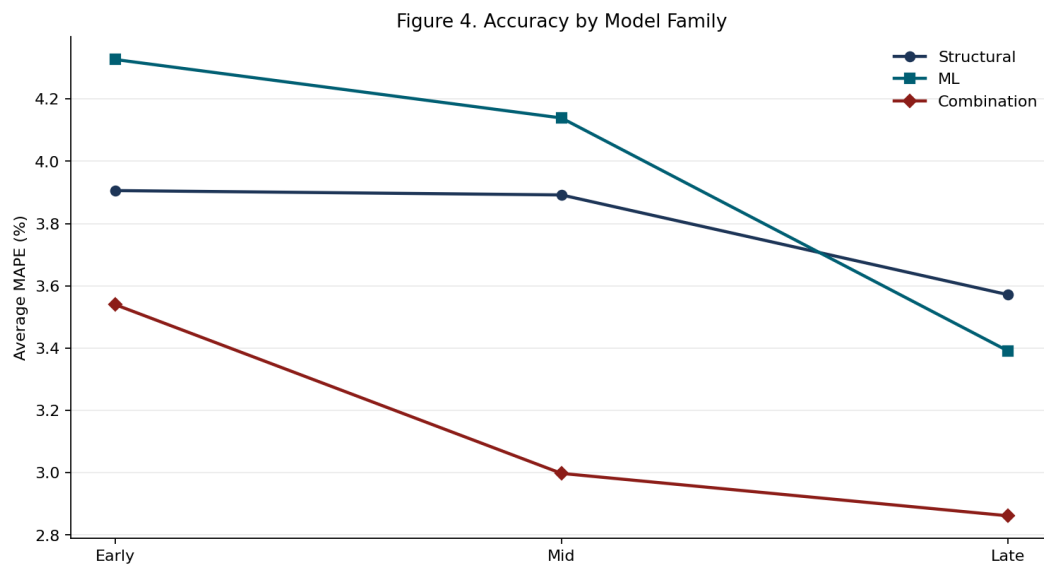


Figure 12: Comparison of structural, machine-learning, and combination model families. The figure shows that combination models dominate on average, which supports the choice of `AdaptiveEnsemble` as the preferred operational specification.

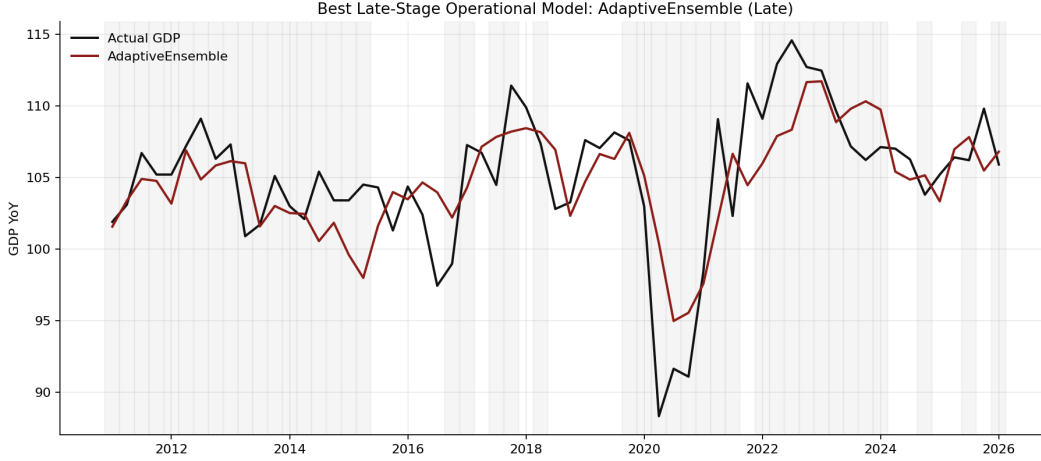


Figure 13: Actual GDP growth and the preferred late-stage operational model. The Adaptive Ensemble tracks the realized GDP series closely while remaining stable during both normal and shock periods.

Figure 9 helps explain why the month-1 crisis case is so difficult. In 2020 Q2 *Early*, the model had to rely on fast market and search-based information because no same-quarter official monthly activity indicators had yet been released.

Figure 10 acts as the methodological bridge between the data and results sections. It explains why the same economy may require different model classes at different points within the same quarter.

Figure 11 adds an important central-bank-style dimension to the analysis. A nowcast is not only a point estimate; it also carries uncertainty. Including that uncertainty makes the empirical discussion more complete.

Figure 12 gives a compact explanation for one of the main empirical findings of the chapter: the best average performer is not the most elegant standalone model, but the combination-model family. Figure 5 complements this with an ensemble-style real-time view. Figures 6 and 7 make the stage dependence of model choice explicit, while Figure 8 shows why robustness in both crisis and non-crisis conditions must be part of the evaluation. Figure 4 adds a direction-of-error interpretation by showing which models overpredict and which underpredict. Finally, Figure 13 provides a direct operational justification for selecting **AdaptiveEnsemble** as the preferred final nowcasting model.

1.7 Final Conclusion for the Thesis

The evidence presented in this chapter leads to four main conclusions.

First, a leakage-safe expanding-window evaluation is essential in a defensible nowcasting study. The quality of the Armenian GDP nowcast changes substantially across information stages, so a single static train-test split is not sufficient.

Second, the best *overall operational model* is **AdaptiveEnsemble**. Its strength comes

from combining model classes that react differently across normal and shock periods.

Third, the best *theory-grounded mixed-frequency benchmark* is the true DFM. Even when it is not the overall winner in average forecast accuracy, it remains the most academically defensible structural model because it handles ragged-edge monthly information directly.

Fourth, the most difficult month-1 crisis case, namely **2020 Q2 Early**, can be handled credibly only with a dedicated shock-adjusted benchmark. The final **EarlyShockAdjusted** specification reduces that error materially and shows that crisis nowcasting in small open economies requires explicit regime-sensitive correction rather than a purely static model choice.

Overall, the contribution of the thesis is not the identification of one universally best model. It is the construction of a *literature-informed, pseudo real-time, shock-aware nowcasting system for Armenia* in which model choice is allowed to vary with the information stage and the crisis regime. An important part of that contribution is the explicit use of alternative data sources: they are most valuable in the *Early* stage, complementary in the *Mid* stage, and still informative in the *Late* stage as rapid indicators of external stress and regime change. That gives the thesis both academic and practical value: it shows how a small open economy can build a transparent nowcasting framework that remains useful in both normal periods and abrupt shock episodes.